AZURE CLOUD DEPLOYMENT TOPOLOGY

POD Management Platform | Next.js 15 | PostgreSQL | Prisma ORM | JWT | 200 Concurrent Users | 2,000 Active PODs | March 2026

CI / CD PIPELINE

GitHub Repository

Source code | main branch | PR-protected

push to main

GitHub Actions

Trigger: push to main

pnpm install

npx prisma migrate deploy

pnpm build --> Deploy to App Service

triggers

reads secrets

GitHub Secrets

AZURE_WEBAPP_PUBLISH_PROFILE

DATABASE_URL (used for migrations)

App Service | Configuration

DATABASE_URL (PostgreSQL + PgBouncer)

JWT_SECRET

NODE_ENV = production

PORT = 3000 | NEXTAUTH_URL

injects env vars

Build & Startup Command

Build: pnpm build

Start: node server.js

deploys code

200 Concurrent Users -- Engineers | Admins | Priority Users | Super Admins

All roles working simultaneously | HTTPS TLS 1.3 | JWT Bearer token on every request | Role-based access enforced server-side

HTTPS

Azure App Service -- Basic B2 | 2 vCPU Dedicated | 3.5 GB RAM | Linux | Node.js 20 LTS | ~$26/month

Dedicated compute -- 100% of 2 vCPU available continuously. No burst credit system.

Single process via node server.js: Next.js 15 SSR + All API Routes + Prisma ORM + JWT Authentication + node-cron scheduler

Excel Export: query DB, build .xlsx in-memory (~88 MB RAM peak per export), stream to browser, memory freed on completion.

Excel Import: xlsx library parses file entirely in the browser. Only JSON payload sent to server. Zero file memory on server.

Notification polling: GET /api/notifications every 30s -- simple indexed SELECT on Notification table.

Next.js 15 SSR | API Routes | Prisma ORM | JWT Auth | node-cron | node server.js

SSL | PgBouncer port 6432 | connection_limit = 10

Database Connection:

postgresql://user:pass@server.postgres.database.azure.com:6432/poddb?sslmode=require&connection_limit=10&pool_timeout=10

PgBouncer

Port 6432 | Built-in | $0

Prisma queries | private VNet | SSL enforced

Azure Region -- East US | App Service and PostgreSQL communicate over private virtual network | PostgreSQL has no public internet endpoint

Azure Database for PostgreSQL -- Flexible Server -- Zone-Redundant High Availability

Burstable B2ms | 2 vCPU | 8 GB RAM | 128 GB SSD | Auto-grow | SSL enforced | Zone-Redundant HA | ~$213/month (compute x2 + storage)

Note: PostgreSQL B2ms is a burstable SKU. Adequate for queries averaging under 5ms. Monitor credit balance after go-live; upgrade to General Purpose tier (+$70/month) if sustained analytics load is detected.

Availability Zone 1 -- PRIMARY

PostgreSQL PRIMARY (Read / Write)

All application reads and writes directed here.

Tables:

Pod | User | Notification | Transaction

PodStatusHistory | LogIssue | AutofillPod

PgBouncer on port 6432.

connection_limit = 10 pooled connections.

Write-Ahead Log (WAL)

Write confirmed to application ONLY after:

1. WAL record written to disk (AZ1).
2. Standby (AZ2) has acknowledged receipt.

If application receives success, data is durable on both zones. Partial writes and silent data loss are impossible.

synchronous streaming replication

auto-failover ~60s | zero data

Availability Zone 2 -- STANDBY

PostgreSQL STANDBY (Hot Replica)

Exact copy of Primary, updated synchronously.

Not used for reads -- zero load, purely on standby.

Automatic failover:

T + 0s : Primary becomes unreachable.

T + 30s : Azure health checks detect failure.

T + 60s : Standby promoted to new Primary.

T + 65s : Prisma reconnects automatically.

Result: Zero data loss. ~60 second recovery.

Automated Backups & Point-in-Time Recovery

Retention: 7 days (configurable to 35 days).

WAL archived continuously. Full daily backup.

Restore to any specific second in the past 7 days.

Included -- no additional charge.

SIZING & COST

App Service -- Basic B2

2 vCPU Dedicated (not burstable)

3.5 GB RAM | Linux | Node.js 20

Handles 200 users comfortably

34 simultaneous Excel exports safe

Fixed pricing, no per-request billing

~$26 / month

PostgreSQL Flexible Server

Burstable B2ms | 2 vCPU | 8 GB

128 GB SSD | Auto-grow enabled

Zone-Redundant HA (billed x2)

Primary compute: ~$99/month

Standby compute: ~$99/month

Storage 128 GB: ~$15/month

Backups: included

PgBouncer: included

~$213 / month

-- Pay-as-you-go Total --

~$239 / month

+ ~$15-30 buffer (bandwidth, monitoring, DNS)

Budget ceiling: ~$270/month

-- 1-Year Reserved (recommended) --

App Service: ~$18/month

PostgreSQL: ~$120/month

Storage/misc: ~$15/month

Target: ~$165/month

Saves ~$1,000 / year

Upgrade path (no redesign):

Raise App Service SKU or PostgreSQL tier independently as user demand grows.

Solid arrow -- live request / query / replication flow

Dashed arrow -- configuration / secret injection at deploy time

WAL guarantee -- write durable on both AZ1 and AZ2 before confirmed

HA failover -- ~60s automatic recovery, zero data loss